

MUHAMMAD NAVEED

Senior Full-Stack Engineer · Technical Lead

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SUMMARY

Full-stack engineer and technical lead with 10 years building, shipping and operating enterprise software. Five of those years in Riyadh on TAMM, a Saudi government digital services platform, leading frontend architecture across multiple feature teams while building the Java/Spring Boot services behind it and owning production issues end to end. Comfortable at both ends of delivery — scoping requirements directly with stakeholders and end users, then carrying the result through containers, CI/CD, cloud infrastructure and post-go-live support. Depth in Angular and TypeScript, breadth across backend, data, cloud and platform operations.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Java, Python, SQL
Frontend	Angular (through v19), RxJS, React Native, HTML5/CSS3, responsive UI
Frontend Arch.	Component architecture and shared design systems, Nx monorepos, micro-frontends / module federation, state management (NgRx, Redux, signals), localisation and Arabic RTL / English LTR, bundle and payload performance
Backend	Spring Boot, Node.js / NestJS, REST & GraphQL API design, WebSockets, microservices, distributed systems, event-driven & asynchronous processing, background jobs
Architecture	Clean Architecture, domain modelling, API contracts, service boundaries, authentication & authorisation (JWT / OAuth2, RBAC, claims-based), API key management, multi-tenant data isolation, open-source platform integration (OpenMRS, Odoo)
Data	PostgreSQL, MySQL, MongoDB, Redis, relational modelling, schema design, query optimisation, caching, Flyway migrations
Cloud & Infra	AWS (ECS/Fargate, ECR, ALB, CloudWatch), Kubernetes, OpenShift, Docker, GCP (GKE, Cloud Run, Cloud SQL), Terraform, Helm, ArgoCD / GitOps
CI/CD & Testing	GitHub Actions, GitLab CI, Azure DevOps, Jenkins, Git, Jest, Karma, Cypress, Playwright, JUnit, Mockito, Testcontainers, unit / integration / end-to-end testing
Reliability	Datadog, Prometheus, Grafana, OpenTelemetry, CloudWatch, structured logging, metrics, tracing, incident management, RCA, versioned releases and rollback
Applied AI	Python LLM services, OpenAI/Claude API integration, streaming responses, automation
Familiar	React, Next.js, RAG, embeddings, vector search, agentic workflows, LLM evaluation — working knowledge, not production experience
Delivery	Agile/Scrum, requirements scoping with stakeholders and end users, technical documentation, UAT, release readiness, go-live and rollout support

EXPERIENCE

Senior Software Engineer (Full Stack) — Elm · TAMM Platform Oct 2021 – Present

Riyadh, Saudi Arabia · National government digital platform — digital services, APIs and government integrations, serving millions of users

- Lead the frontend architecture for TAMM — component model, shared design system, Nx monorepo with micro-frontend boundaries, state management patterns and the delivery standards feature teams build against.
- Deliver bilingual Arabic/English service journeys end to end in Angular/TypeScript — RTL/LTR layout, API integration, validation, state management (NgRx / signals) and error handling.
- Design and build the Java/Spring Boot services and REST APIs behind those journeys in a distributed microservices environment, using asynchronous and event-driven patterns where integration boundaries call for them.
- Scope requirements directly with government and business stakeholders, then turn them into technical designs, service boundaries and delivery plans.
- Own the technical calls on API design, data flow, caching, failure handling and release readiness, and carry them to agreement with architects, analysts, QA, security and infrastructure.
- Rebuilt CI/CD across product teams on GitHub Actions and GitLab CI, cutting deployment time by ~30%; set the testing standard with Jest/Karma unit coverage and Cypress/Playwright end-to-end suites, and carry releases through UAT, deployment verification and post-release monitoring.
- Operate production workloads on AWS, Kubernetes and OpenShift with Datadog observability, and designed the platform's authentication and access model — JWT/OAuth2, RBAC and API key management for third-party integrations.
- Lead production incident investigation across frontends, APIs, databases, Redis and infrastructure — root cause from logs, metrics and runtime behaviour through to the fix.
- Improved performance by 20–25% on a high-traffic citizen service journey after profiling — bundle size and lazy loading, query optimisation, payload shaping and caching — alongside structured logging and clearer runtime diagnostics.
- Mentor junior and mid-level engineers through design and code review, refactoring and testing, and unblock issues that cross service or team boundaries.

Full-Stack Engineer (Java & Node.js) — LMKR · Halliburton Digital Well Program Sep 2019 – Sep 2021

Islamabad, Pakistan · Enterprise engineering software for oil-and-gas workflows, international teams and large engineering datasets

- Built Angular/TypeScript frontend modules for data-intensive engineering workflows, with reusable components, validation and usability as the focus over large engineering datasets.
- Worked across Java/Spring Boot services and a Node.js codebase, designing versioned REST and GraphQL APIs so evolving the system never broke existing clients, with long-running operations, asynchronous processing and background jobs.
- Engineered real-time well monitoring over WebSocket streams with retry and exponential backoff, built for unstable field networks where naive implementations fail.
- Scoped specialist oil-and-gas workflows directly with domain experts and international product teams, and turned them into concrete interfaces, APIs and data flows.
- Consolidated scattered backend calls into a single aggregation layer, cutting API response times by ~15%; improved query and data-access performance on large engineering datasets by reworking SQL and access patterns.
- Contributed to system design, API contracts, service boundaries and reusable implementation patterns; used Docker and Jenkins for CI, release support and deployment troubleshooting.

Islamabad, Pakistan · Healthcare and ERP products built on open-source platforms — OpenMRS and Odoo

- › Built and extended the frontend of an OpenMRS-based electronic medical record system — patient registration, clinical encounter forms, observation capture and reporting views — against the OpenMRS REST API and its patient / encounter / observation / concept data model.
- › Developed custom OpenMRS UI modules on top of the platform's module system, so clinic-specific workflows could be added without forking or patching the core platform.
- › Built reusable Angular and JavaScript components and responsive, data-driven interfaces over REST APIs, with cross-browser support across the product suite.
- › Customised Odoo frontend views and workflows (QWeb templates, form/list views) for billing, stock and reporting, and connected them to external systems over Odoo's XML-RPC and REST interfaces.
- › Built the integration between OpenMRS and Odoo so clinical activity flowed through to billing, inventory and procurement — mapping patients, services and stock items across two different domain models and reconciling their identifiers.
- › Handled the integration failure modes — mismatched data models, ID mapping, synchronisation ordering, partial failures and retries — and surfaced failed records so clinic staff could resolve mismatches themselves instead of raising tickets.
- › Deployed and configured the product in client environments, scoped requirements on site with clinic and operations staff, and ran onboarding, data migration and go-live support.

INDEPENDENT PROJECTS

SponsorOne — Sponsorship & Donation Platform for Education NGOs

Live in production

Java 17 · Spring Boot 3 · PostgreSQL · Angular · Docker · Flyway · Testcontainers | AWS ECS/Fargate · Heroku · Terraform · GKE · Helm · ArgoCD

- › Sole engineer from business requirements through architecture, build, go-live and ongoing operation — sponsorone.app, live in production.
- › Framework-free domain layer following Clean Architecture, with immutability enforced in both the domain code and the PostgreSQL schema; append-only financial ledger with claims-scoped access and organisation scoping at every table, giving per-tenant isolation and a system-wide audit trail.
- › Designed the PostgreSQL data model and migration strategy, with automated and integration tests (Testcontainers) around the behaviour that matters.
- › Run deliberately across vendors to build platform depth on a system with real users and real financial records: AWS ECS/Fargate in production first, Heroku after a cost-driven move, and a Terraform-provisioned GKE build-out with Helm, ArgoCD/GitOps and Prometheus/Grafana/OpenTelemetry underway. The framework-free domain layer is what made each move a deployment change rather than a rewrite.

Spreadlee — Production AWS Deployment & Operations

Docker · Amazon ECR · ECS/Fargate · Application Load Balancer · CloudWatch · Redis

- › Containerised the application and ran production deployments through ECR into ECS/Fargate behind an Application Load Balancer.
- › Configured health checks and CloudWatch monitoring; diagnosed and resolved Redis issues affecting live traffic.
- › Operated versioned releases with a working rollback path, so a bad deploy was a recoverable event rather than an incident.

EDUCATION

BS, Computer Science — University Institute of Information Technology (PMAS-AAUR), Rawalpindi, Pakistan

Sep 2016

English — professional working proficiency · Urdu — native · **Saudi Iqama — transferable** · Open to roles across Saudi Arabia and the wider GCC